Agentic Flow

The Active-Pilot Support System

Problem Statement : AIAG01–AI-Driven Website-Integrated Business Support System

Team Name : VOYAGERS

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Problem & Motivation

CURRENT PROBLEMS

- Traditional chatbots are "Passive Text Generators." They can answer basic FAQs but cannot do anything on the website (like show you a product).
- The "Context Gap": When a bot fails, the user is transferred to a human who has zero context, forcing the user to repeat their anger-inducing story.

IMPORTANCE

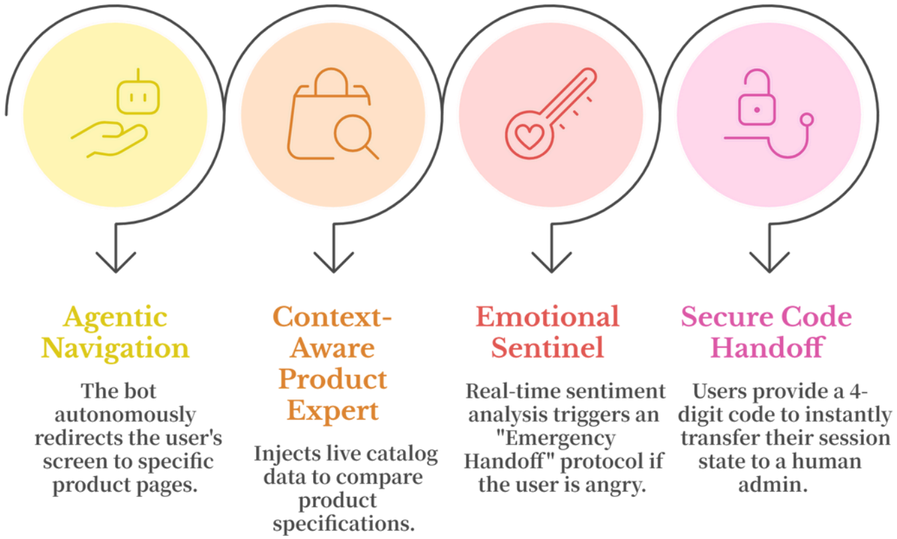
- High Abandonment: Users leave when they can't find products quickly.
- Operational Waste: Humans spend 40% of support time just collecting basic details (Order ID, issue) that the bot should have handled.

GAPS IN CURRENT SOLUTIONS:

- Isolated widgets that don't control the website UI.
- "Dumb" Handoffs (e.g., just emailing a ticket).
- Lack of emotional intelligence (treating angry users the same as curious ones).

Solution & Key Features

We built an Agentic AI System that acts as an "**Active Pilot**" for the website. It doesn't just talk; it navigates the user to products, reads live order data, and detects sentiment to trigger a secure, context-rich human handoff.



Key Features:

- 1. Agentic Navigation: The bot autonomously redirects the user's screen to specific product pages (e.g., "Show me the drone" -> Site Auto-Navigates).
- 2. Context-Aware Product Expert: Injects live catalog data to compare specs (e.g., "Which laptop has better RAM?").
- 3. Emotional Sentinel: Real-time sentiment analysis that triggers an "Emergency Handoff" protocol if the user is angry.
- 4. Secure Code Handoff: A novel, privacy-first mechanism where users provide a 4-digit code to instantly transfer their session state to a human admin.

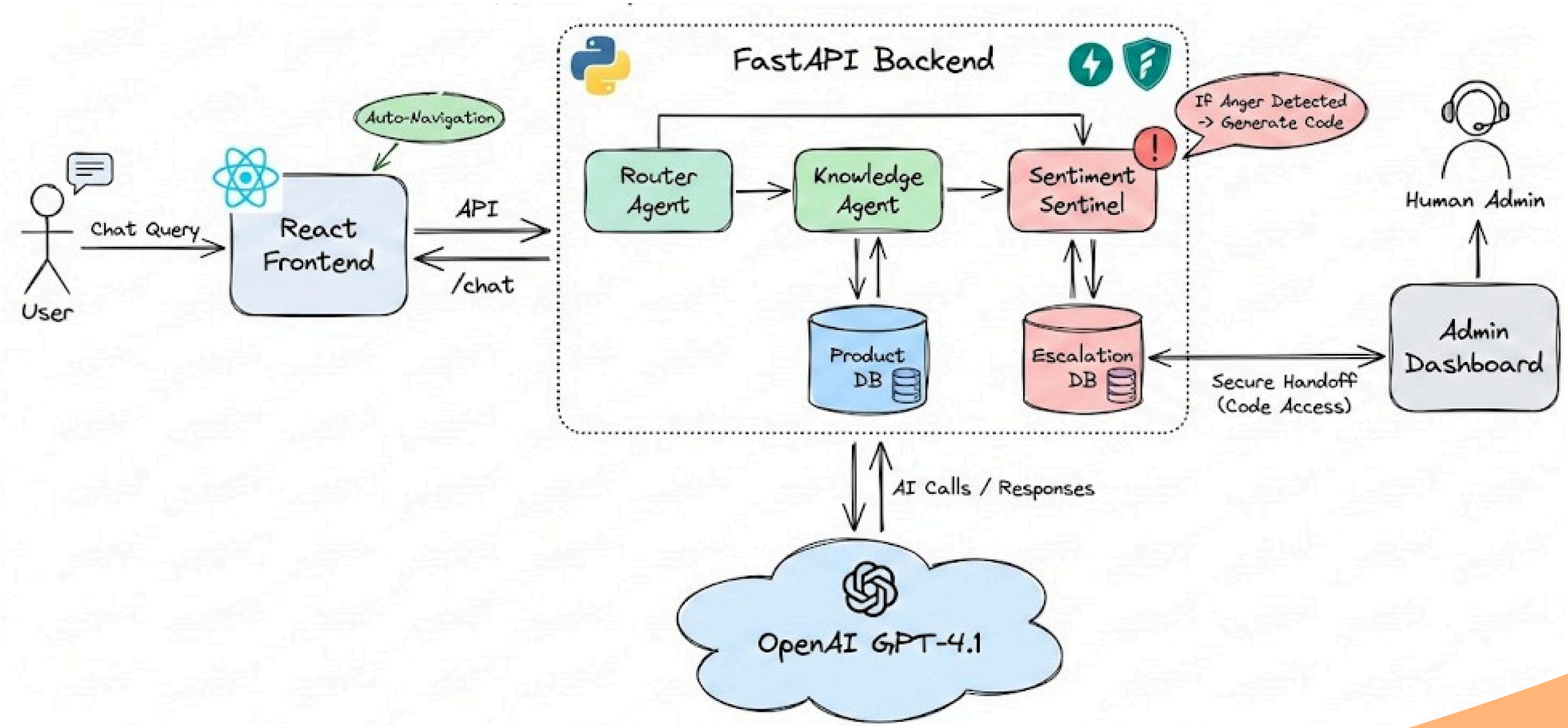
What makes the solution unique?

- Most bots are "Text-Only." Ours is "Multi-Modal Action"—it controls the browser URL and UI state based on conversation intent.

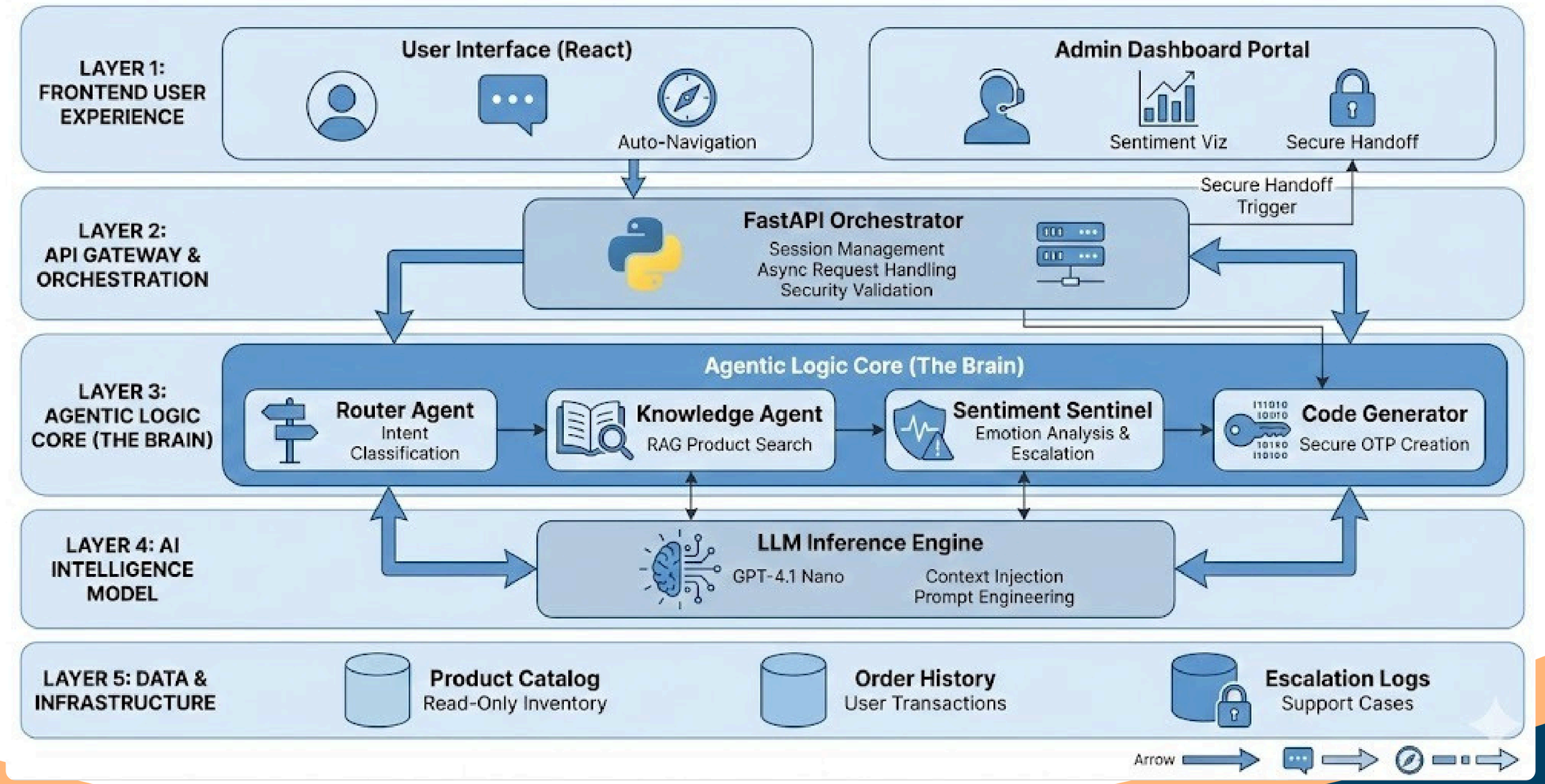
TECH STACK USED

Category	Technology
Frontend	React.js (Vite), Tailwind CSS, Lucide React
Backend	Python (FastAPI), Uvicorn Server
Database	In-Memory JSON Store
Frameworks	React Router DOM (Navigation), Pydantic (Validation)
APIs / Integrations	OpenAI API (GPT-4.1 Nano), RESTful API
Cloud / Hosting	Localhost (Dev), Vercel (Frontend Ready), Render (Backend Ready)
Devops / CI - CD	Git, GitHub
Authentication	Custom Secure OTP
Tools and Platforms	VS Code, Postman, Chrome DevTools

WORKFLOW


FastAPI Backend
Auto-Navigation
User
Chat Query
React Frontend
API
/chat
Router Agent
Knowledge Agent
Sentiment Sentinel
If Anger Detected -> Generate Code
Human Admin
Product DB
Escalation DB
Secure Handoff (Code Access)
Admin Dashboard
AI Calls / Responses
OpenAI GPT-4.1

ARCHITECTURE



Achievements & Impact

Features

E-Commerce Store
A fully functional mock e-commerce store has been completed.

Stateful Chatbot
A stateful chatbot with memory has been implemented.

Autonomous Navigation
Autonomous page navigation has been successfully integrated.

Admin Dashboard
An admin dashboard with sentiment analysis visualization is ready.

Impact

Zero Context Loss	Proactive Resolution
Exact conversation history instantly	Bot offers solutions before humans
Support agents see history instantly	Bot offers solutions before humans
Support agents see history instantly	Bot offers solutions before humans

Future improvements

Voice Integration
Users can speak the handoff code instead of typing it.

Multi-Agent Swarm
Separate agents for "Technical Support" and "Sales" work together.

Real Database
Migrating from JSON to PostgreSQL for production scale.

Thank You